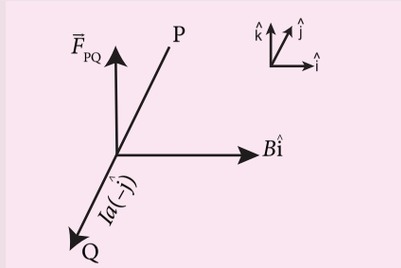
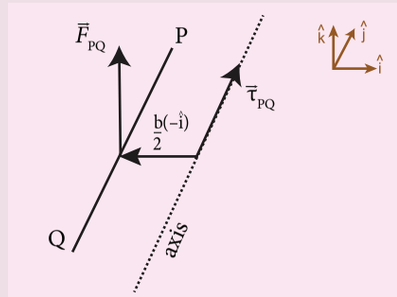
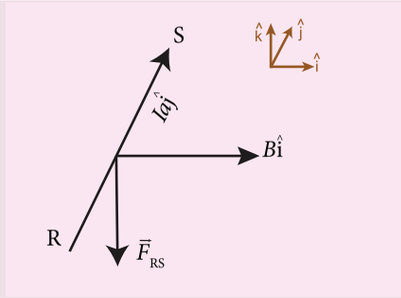
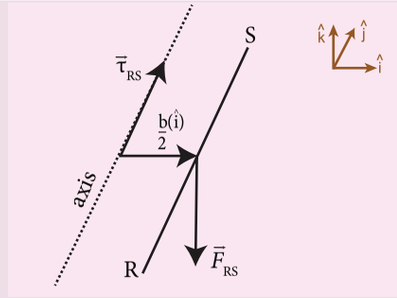
Appendix-1.indd 321

Table A3.1 Unit normal vector is perpendicular to magnetic field

Section	Force		Torque	
	$\vec{F} = I\vec{l} \times \vec{B}$	Explanatory diagram	$\vec{\tau} = \vec{r} \times \vec{F}$	Explanatory diagram
PQ	$\vec{F}_{PQ} = I(-a\hat{j}) \times B\hat{i}$ $= -IaB(\hat{j} \times \hat{i})$ $= IaB\hat{k}$		$\vec{\tau}_{PQ} = \frac{b}{2}(-\hat{i}) \times \vec{F}_{PQ}$ $= \left(-\frac{b}{2}\right)(IaB)(\hat{i} \times \hat{k})$ $= \frac{b}{2} IaB\hat{j}$	
QR	$\vec{F}_{QR} = Ib\hat{i} \times B\hat{i} = \vec{0}$		$\vec{\tau}_{QR} = \frac{a}{2}(-\hat{j}) \times \vec{F}_{QR} = \vec{0}$	
RS	$\vec{F}_{RS} = (Ia\hat{j}) \times B\hat{i}$ $= IaB(\hat{j} \times \hat{i})$ $= -IaB\hat{k}$		$\vec{\tau}_{RS} = \frac{b}{2}\hat{i} \times \vec{F}_{RS}$ $= \frac{b}{2}\hat{i} \times (-IaB\hat{k})$ $= \left(-\frac{b}{2}\right) IaB(\hat{i} \times \hat{k})$ $= \frac{b}{2} IaB\hat{j}$	
SP	$\vec{F}_{SP} = Ib(-\hat{i}) \times B\hat{i} = \vec{0}$		$\vec{\tau}_{SP} = \frac{a}{2}\hat{j} \times \vec{F}_{SP} = \vec{0}$	

23-12-2021 19:25:45